Modeling Competitive Advantage Through Multi-Agent Reinforcement Learning in Virtual Markets

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Abstract. This study presents an AI-powered strategy simulator that can demonstrate significant empirical capacity to model competitive market dynamics using multi-agent reinforcement learning (MARL). Moreover, the agents in this system show that firms could demonstrate complex interactive behavior within these important theoretical conditions. Thus, agents might show sequential decisions to affect innovation, pricing, and market participation. The system provides a fully simulated virtual environment for economic activity. The framework integrates several factors that affect competitive conditions. These factors include Porter's Five Forces, STEP environmental factors, complements, and cost-structure modeling. The integration of these factors allows the system to show patterns in how advantages in competition develop. The development occurs under conditions that change over time, where RL comes into the picture. Through this repeated interaction, the agents develop strategies that approach optimal forms. The strategies that develop affect several outcomes. These outcomes include the intensity of competition between firms, the reasons for firms to develop new approaches, and the level of profit across the industry. The framework provides a critical analytical platform, which can demonstrate important capacity for examining behavior relating to strategy, allowing for assessment of economic profit in markets that show complex features and change in response to conditions.

Keywords: multi-agent reinforcement learning (MARL), reinforcement learning (RL), competitive strategy, economic profit, business simulation, adaptive markets.

1. Introduction

Research on competitive strategy and industrial organizations may well suggest how firms sustain differential performance in concentrated industries. Moreover, classical frameworks such as Porter's Five Forces might indicate that structural determinants of profitability including rivalry intensity, buyer power, supplier power, threat of substitutes, and barriers to entry (Porter, 2008) demonstrate that these critical factors substantially influence outcomes. Furthermore, complementary perspectives such as the Resource-Based View may suggest that important theoretical evidence supports firm heterogeneity and persistent advantage rooted in accumulated capabilities (Barney, 1991). Thus, industrial organizations could show formal models of price competition that affect endogenous market structure. However, research may show Bertrand competition extends to differentiated product demand systems and dynamic innovation models (Berry, Levinsohn, & Pakes, 1995; Sutton, 1991; Aghion et al., 2001). Many traditional models may well suggest that strategic interaction could be examined through significant equilibrium solution concepts that impose rational expectations and fixed strategy spaces. Moreover, the important evidence might indicate that dynamic environments characterized by technological change, macroeconomic volatility, and regulatory constraints make solving Markov Perfect Equilibria analytically complex. Thus, such approaches could require restrictive assumptions. However, strategy frameworks may show analytical limits. Additionally, frameworks such as Porter's Five Forces or STEP analysis might not model how firms adapt sequentially in response to evolving market states.

Recent advances in multi-agent reinforcement learning (MARL) may well suggest that decentralized learning approaches provide significant empirical support for understanding how coordinated outcomes emerge across repeated competitive interactions. Considering the important theoretical findings, decentralized learning agents might indicate that pricing strategies could demonstrate substantial adaptation over time, potentially converging toward coordinated outcomes without explicit communication (Calvano et al., 2020; Klein, 2021). Moreover, the evidence may suggest that these results support renewed interest in algorithmic collusion. Thus, findings might show

dynamics of tacit coordination affecting market behavior. However, much research could indicate that this literature employs stylized environments with simplified demand and cost structures, abstracting from endogenous innovation, regulatory frictions, and macroeconomic shocks.

This paper may well suggest that a significant computational industrial organization framework could demonstrate that three firms compete in prices and R&D within a structurally specified oligopolistic market. Moreover, the model appears to indicate that a differentiated Bertrand demand system consistent with empirical IO practice (Berry et al., 1995) supports convex innovation costs, cumulative innovation dynamics, regulatory price constraints, supplier cost volatility, and macroeconomic regime switching. Furthermore, the evidence may suggest that innovation enhances product attractiveness and generates path-dependent competitive advantage, linking IO models of endogenous concentration (Sutton, 1991) with strategy concepts of sustained competitive advantage (Barney, 1991). Thus, firms may maximize discounted intertemporal profits and adapt through decentralized Proximal Policy Optimization (PPO). However, results might show competitive patterns emerge from repeated interaction in a fully observable stochastic environment. Given that the significant empirical findings could indicate that this critical framework substantially operationalizes structural features, the important theoretical results may well demonstrate that industry forces, cost drivers, and substitute pressure exist within a quantitatively explicit IO model. Additionally, the key evidence might suggest that competitive patterns pricing behavior, concentration, and innovation races could demonstrate relevant path-dependent outcomes. Moreover, results may show equilibrium conditions need not be imposed. Considering these findings, research could indicate frameworks allow concentration to emerge naturally. Nevertheless, data might show strategy research benefits from quantitatively explicit models. The results and analysis address three main questions. First, do firms converge pricing outcomes consistent with differentiated Bertrand competition, or do they approach highly competitive regions that are suggestive of tacit coordination? Second, when there is a fixed market situation, then is just investing in innovation that can make them leader? Third, how do external factor shocks and regulatory restrictions shape the strategic responses of the agents? Our contribution is made up of incorporating MARL and structural IO modeling yet keeping a clear economic interpretation of every parameter and mechanism. In addition, the framework offers a controlled lab to investigate innovation-driven advantage, natural concentration, and repeated-game pricing behavior under practical limitations. The study develops computational techniques for examining dynamic oligopoly behavior by tying classical IO theory with adaptive learning dynamics.

2. Literature Review

This paper contributes to four streams of scholarship:

- Differentiated oligopoly and structural industrial organization.
- Endogenous innovation and dynamic competition.
- Algorithmic pricing and tacit collusion.
- Multi-agent reinforcement learning in economic environments.

2.1 Differentiated Oligopoly and Structural Industrial Organization

The modern theory of oligopoly dates to Bertrand (1883), who showed that homogeneous price competition results in marginal cost pricing under standard assumptions. Later work showed that product differentiation is a fundamentally different concept that changes the outcome and allows for the existence of equilibrium markups. Structural models of demand estimation, especially the discrete choice model developed by Berry, Levinsohn, and Pakes (1995), formalized the role of substitution patterns in determining pricing power in differentiated industries.

In such models, equilibrium markups depend on own-price elasticity. Under standard conditions, the Lerner index implies:

$$\frac{P_i - C_i}{P_i} = -\frac{1}{\varepsilon_i}$$

(equation 1)

where ε i denotes the firm's own-price elasticity of demand. Markups may well suggest that significant empirical factors including how customer demand behaves and how close competitors are could plausibly demonstrate their critical influence on the observed outcomes across markets. In light of the important findings from a substantial body of research, the evidence might reasonably indicate that these key determinants appear to shape our understanding of how markets work. Furthermore, the results may suggest that this theoretical framework influences how demand is modeled. However, findings might indicate a basic structure affecting market outcomes. Thus, research may show concentration shapes of how firms compete. The idea of endogenous market structure could indicate that significant theoretical implications follow strategic firm behavior rather than consolidation alone. Moreover, the important evidence may suggest that market concentration when a few firms dominate might demonstrate that strategic investments drive outcomes. Given that (Sutton ,1991) showed that in industries where companies spend heavily on quality and other sunk costs, the market often stays concentrated even without mergers, the findings could demonstrate that this critical sunk cost factors substantially influence observed market structure. Nevertheless, evidence may show sunk costs affect concentration. Therefore, research might indicate investments drive structure. Considering the significant empirical findings that further support this idea, the results could indicate that Autor et al. (2020) found that "superstar firms" tend to emerge in such markets. Additionally, the important evidence may suggest that these firms become dominant because they demonstrate that productivity and superior technology influence competitive outcomes. However, findings may show dominance affects market share. Thus, research might indicate productivity shapes firm outcomes. De Loecker, Eeckhout, and Unger (2020) demonstrate that markups have increased significantly across industries, suggesting increased concentration driven by structural asymmetries.

Given that the significant theoretical contributions continue to develop, dynamic models of industrial organization could indicate that these important analytical approaches may well extend this line of thinking. However, (Ericson and Pakes (1995)) may suggest that industry dynamics analyzed using Markov perfect equilibrium could demonstrate how investment, learning, and exit drive persistent heterogeneity. Additionally, research might show computational tools affect dynamic oligopoly analysis. In light of these findings, Doraszelski and Pakes (2007) and Pakes and McGuire (2001) may show results support broader analytical development. Therefore, the current research may continue this line but substitutes equilibrium solution methods with decentralized adaptive learning, permitting competitive outcomes to emerge endogenously from interaction.

2.2 Endogenous Innovation and Dynamic Competition

Innovation-driven rivalry could indicate that both industrial organization and strategic management share fundamental theoretical foundations. Given that Aghion et al. (2001) formalize step-by-step innovation models in which firms invest to climb a quality ladder, the significant empirical framework may well suggest that two critical mechanisms emerge: firms innovate to escape competition, while incumbents invest to preempt entry. Thus, innovation may shape competitive intensity and market structure. However, these findings might affect how scholars examine persistent leadership. Moreover, recent work may show heterogeneous innovation drives durable dominance.

In light of the significant empirical evidence, Akcigit and Kerr (2018) demonstrate that innovation capabilities vary systematically across firms and may contribute to important asymmetric outcomes over time. Furthermore, the key findings suggest that R&D-intensive industries could exhibit stable asymmetries without requiring predatory behavior. However, strategy perspectives may show these dynamics intersect with established theory. Additionally, findings might indicate resource-based views matter. Notwithstanding these results, dynamic capabilities could affect how firms sustain advantage. The resource-based view (Barney, 1991) and dynamic capabilities theory (Teece, 2018) may well suggest that sustained advantage arises from cumulative capability development rather than short-term tactical positioning, a finding the significant theoretical evidence could indicate deserves careful consideration. Moreover, the important results demonstrate that Cockburn, Henderson, and Stern (2018) suggest that artificial intelligence technologies may amplify such capability asymmetries by enhancing firms' ability to learn and adapt. Therefore, evidence may show accumulation matters more than pricing. Thus, findings might indicate exclusionary behavior appears less relevant. Nevertheless, capability development could shape market outcomes.

Given that the significant theoretical and empirical evidence could plausibly demonstrate that concentration emerges from capability accumulation rather than exclusionary pricing, the present framework may well suggest that these important insights operate within a dynamic oligopolistic environment in which innovation accumulates over time and shifts demand attractiveness. Additionally, the key results might indicate that operationalizing these findings reveals critical structural patterns. However, evidence may show demand attractiveness shifts concentration. Furthermore, findings might affect how firms accumulate capabilities. Moreover, results could indicate market structure emerges naturally.

2.3 Algorithmic Pricing and Tacit Collusion

The diffusion of algorithmic pricing has renewed concerns regarding tacit coordination. Traditional repeated-game theory shows that supracompetitive outcomes can be sustained without explicit communication under sufficient observability (Fudenberg & Levine, 1998). Recent computational research extends this logic to reinforcement learning environments.

Calvano et al. (2020) demonstrate that independent Q-learning agents in a repeated Bertrand game can converge to collusive pricing even without explicit coordination. Klein (2021) reports similar dynamics in sequential pricing settings. Harrington (2018) and Ezrachi and Stucke (2016) discuss the regulatory implications of algorithmic environments that may facilitate coordination beyond traditional detection mechanisms.

However, most algorithmic collusion studies rely on stylized homogeneous-good demand systems and abstract from innovation investment, regulatory constraints, and cost volatility. Consequently, their applicability to innovation-intensive, regulated industries remains limited. By embedding learning agents within a differentiated-demand structure featuring innovation accumulation and institutional constraints, the present study evaluates whether adaptive pricing yields collusive outcomes or disciplined competition under structurally richer conditions.

2.4 Multi-Agent Reinforcement Learning in Economic Environments

Reinforcement learning models strategic interaction as repeated decision-making under uncertainty, where agents maximize discounted returns:

$$V_i(s) = E\sum_{t=0}^{\infty} \quad \left[\delta^t \pi_i \quad (t) \;\; s_0 = s\right]$$

(equation 2)

where δ captures intertemporal orientation. In multi-agent environments, the strategic setting becomes non-stationary because each agent's policy influences the state transitions faced by others (Shoham & Leyton-Brown, 2008).

Recent advances in multi-agent reinforcement learning (Busoniu, Babuska, & De Schutter, 2008; Zhang, Yang, & Başar, 2021) demonstrate the capacity of decentralized learners to adapt in complex environments. Yet integration with structurally interpretable economic models remains limited. Rather than solving for Markov Perfect Equilibria analytically (Ericson & Pakes, 1995), decentralized learning can serve as an equilibrium selection mechanism, allowing competitive regularities to emerge from repeated adaptation. This approach is particularly useful in settings characterized by stochastic shocks, innovation accumulation, and regulatory constraints.

The novelty of the present study lies in combining a differentiated demand system, cumulative innovation dynamics, institutional frictions, and decentralized PPO learning within a unified framework. The model preserves economic interpretability while allowing equilibrium patterns to arise from adaptive interaction.

2.5. Research Gap

Innovation theory discusses dynamic leadership and path dependence, structural industrial organization theory illustrates differentiated pricing and endogenous concentration, and algorithmic pricing studies demonstrate how autonomous learning can result in coordinated outcomes. Competitive positioning and the creation of cumulative capability are the main topics of strategy research.

These academic collections, however, are mostly separate. It is still difficult to create an integrated framework that takes into account autonomous adaptive learning, innovation accumulation, regulatory frictions, and structural economic building blocks. By implementing multi-agent reinforcement learning into a comprehensive oligopolistic model, this study closes this research gap. These results are produced endogenously through repeated strategic interactions, as opposed to earlier equilibrium models that impose pricing discipline, innovation leadership, and concentration ex ante.

3.Economic Model

This section may well suggest that a structured model of significant oligopoly competition could plausibly demonstrate the critical interactions between differentiated Bertrand pricing, firms' own innovation investment, regulatory limits, and random shocks in a repeated setting. Moreover, the model might indicate that these important theoretical aims could support keeping the analysis economically clear while allowing significant market outcomes to arise naturally from firms' decisions. Thus, findings may show the model builds on differentiated demand systems in empirical industrial organizations. However, results could indicate endogenous market structure theory (Sutton, 1991) and step-by-step innovation models (Aghion et al., 2001) may affect outcomes. Furthermore, evidence may support that (Berry, Levinsohn, & Pakes, 1995) remains relevant. Considering the significant empirical findings, the model could indicate that three firms $i \in \{1,2,3\}$ might reasonably demonstrate important competitive behavior across discrete time periods $t=0,1,\ldots,T$. Additionally, the key evidence may suggest that firms simultaneously choosing price $P_i(t)$ and innovation investment $R_i(t)$ could demonstrate that forward-looking decisions appear to support maximizing discounted expected profits. Therefore, results may show firms examine prices simultaneously. Notwithstanding this result, data could indicate investment decisions affect profit outcomes. Given that evidence supports this, findings may show firms aim to maximize returns:

$$V_i = E\sum_{t=0}^{T} \left[\delta^t \pi_i \quad (t)\right] \ , \ \delta \in (0,1)$$

(equation 3)

The discount factor δ models' intertemporal orientation. Higher values induce greater emphasis on capability accumulation relative to short-run markup extraction.

3.1. Profit Structure

Per-period profit is defined as:

$\pi_i(t) = P_i(t) \cdot Q_i(t) - C_m(t) \cdot Q_i(t) - kR_i(t)^2 - C_{capital} \quad - \left(C_{fixed} + \tau Q_i(t)\right)$	(equation 4)

Revenue could indicate that price multiplied by the significant empirical quantity sold produces the critical total output measure, with marginal cost Cm (t) appearing to suggest that random temporal variation may well capture important supply-side shocks across the relevant period. Moreover, the quadratic R&D cost term kRi (t)2 may suggest that innovation demonstrates that investment becomes more expensive as firms allocate resources more intensively. Given that the evidence indicates balanced interior levels of investment, the significant findings could demonstrate that innovation costs rise at an increasing rate (Aghion et al., 2001). Thus, fixed capital cost Ccapital might show sunk costs affect entry and operation. However, endogenous concentration models may support this result (Sutton, 1991). Furthermore, regulation may show fixed cost Cfixed and variable compliance cost τQi (t), affect firms. Therefore, the regulatory burden could rise as production increases.

This structure embeds three fundamental trade-offs:

- Markup vs. demand contraction
- Short-run profit vs. long-run innovation advantage
- Scale expansion vs. regulatory exposure

3.2. Aggregate Demand

Industry demand evolves in response to persistent macroeconomic regimes and collective pricing behavior. Baseline demand is given by:

$D_{base}(t) = D_0 \cdot \theta(t)$	(equation 5)

where D_0 steady-state market potential and Θ(t)∈{θB ,θR} represents the macroeconomic state at time t. The economy follows a two-state Markov process:

$$P(\theta_{t+1} = \theta_B | \theta_t = \theta_B) = p_{BB}$$

$$P\left(\theta_{t+1} = \theta_R \; \theta_t = \theta_R\right) = p_{RR} \qquad \text{(equation 6)}$$

with the probability of remaining in a recession regime (PRR) could indicate that booms and recessions may well suggest significant empirical persistence rather than rapid transitional shifts between states. In the model, $\theta B>1$ represents higher demand during booms, while $\theta R<1$ captures lower demand during recessions. Moreover, the important cyclical evidence may suggest that these systematic demand changes demonstrate that lasting regime effects influence overall market conditions. Thus, findings might indicate firms face uncertainty affecting market size. However, firms may show strategic responses to regime continuation. Given that the key evidence demonstrates that effective demand depends on price elasticity and competitive pressure from substitute products, the results could indicate that these critical macroeconomic factors substantially influence the important strategic decisions firms must consider:

:

$$D_{effective}(t) = D_{base} \quad (t) \cdot \exp\left(-\varepsilon P_{avg} \quad (t)\right) \cdot \left(1 - SubstitutePressure(t)\right) \qquad \text{(equation 7)}$$

The parameter ε > 0 appears to demonstrate that overall demand could indicate significant sensitivity to the industry's average price level. Moreover, the important empirical evidence may suggest that when all firms raise their prices, the results demonstrate that total market demand falls. Thus, findings may show this reflects industry-level price elasticity, consistent with differentiated demand systems (Berry, Levinsohn, & Pakes, 1995). However, SubstitutePressure(t) might indicate outside competition, such as generics, imports, or related technologies, could affect total demand. Therefore, evidence may show this reduces demand independently of firms' own pricing strategies. Overall demand could plausibly demonstrate that it appears subject to three critical key forces that substantially influence the observed outcomes.:

1. Persistent macroeconomic cycles (level shifts),
2. Collective pricing behavior (slope effects via elasticity),
3. External substitute competition (exogenous leakage).

This structure implies that pricing decisions influence both relative market shares and total market size-a key departure from models with fixed aggregate demand-and that equilibrium behavior must be robust to cyclical fluctuations rather than stationary conditions.

3.3. Market Share Allocation

Conditional on effective demand, firm-level shares follow a logit specification:

$$S_i(t) = \frac{exp\left(-\alpha P_i(t) + \beta(t) I_i(t)\right)}{\sum_{j=1}^{3} exp\left(-\alpha P_j(t) + \beta(t) I_j(t)\right)} \qquad \text{(equation 8)}$$

The parameter α>0 influences price competition intensity, while Ii(t) represents cumulative innovation stock. The function β(t) determines the strength of innovation in shaping product uniqueness.

Quantity sold is:

$Q_i(t) = S_i(t) \cdot D_{effective}(t).$	(equation 9)

This structure captures differentiated Bertrand competition prices influence relative attractiveness, but innovation shifts demand weights and relaxes direct price rivalry. Markups therefore depend on both own-price elasticity and the degree of technological differentiation (Berry et al., 1995).

3.4. Innovation Dynamics

Innovation evolves cumulatively:

$$I_i\,(t+1) = I_i(t) + R_i(t).$$

(equation 10)

This equation introduces path dependence. Firms that invest early build an early advantage, consistent with stepwise innovation races (Aghion et al., 2001). Innovation effectiveness evolves as:

$$\beta(t) = \beta_0 \quad (1 + 0.002t)\Big(1 + 0.01I_{total} \quad (t)\Big)^{-1}.$$

(equation 11)

Two forces operate simultaneously:

- Exogenous technological progress increases the leverage of innovation over time.
- Industry-wide innovation reduces marginal differentiation, preventing explosive monopolization.

Thus, market concentration can be built up naturally from firms' own strategies, but diminishing returns prevent monopoly.

3.5. Cost Shocks and Regulatory Constraints

Marginal production cost follows:

$$C_m(t) = c^{-} \cdot SupplierShock(t),$$

(equation 12)

where SupplierShock is lognormally distributed. This captures upstream volatility and bargaining dynamics. Pricing is bounded by:

$$C_m(t) + \epsilon \leq P_i \quad (t) \leq P^{-}.$$

(equation 13)

The lower price limit may well suggest that firms face significant empirical constraints against predatory pricing strategies, while the upper limit represents critical regulatory price caps that could plausibly demonstrate the boundaries of permissible competition. Moreover, the evidence might indicate that these rules define the range in which firms can compete. However, rules may not directly decide on final outcomes. Thus, concentration might affect market structure through factors beyond simple regulatory limits. Given that findings reveal long-term differences in innovation and pricing strategies between firms, market concentration could indicate endogenous development rather than external mergers or consolidation, and this may well suggest that the important theoretical results appear consistent with endogenous market structure theory (Sutton, 1991). Furthermore, the significant empirical findings might indicate that the Herfindahl-Hirschman Index demonstrates that concentration is key evidence of structural outcomes:

$$HHI(t) = \sum_{i=1}^{3} S_i(t)^2 .$$

(equation 14)

Results may suggest factors influence competitive dynamics. Nevertheless, data might show structure develops endogenously. Considering these results, evidence may indicate these supports (Sutton, 1991)

4. Computational Framework

Figure 1 may well suggest that the significant empirical system structure demonstrates that the structural economic model demand, innovation dynamics, cost shocks, and regulatory constraints defines the critical environment in which firms could plausibly operate. Moreover, the important theoretical findings indicate that three independent PPO agents might reasonably examine the market conditions and simultaneously choose their price and R&D decisions. Thus, results may show the environment updates of quantities sold, profits, and innovation levels. Additionally, evidence could indicate changes in macroeconomic regimes and supplier-side shocks to emerge, sending rewards back to each agent. Given that these key findings appear relevant, each agent might demonstrate policy updates independently during training, while the policies are fixed and no longer updated during tournament evaluation.

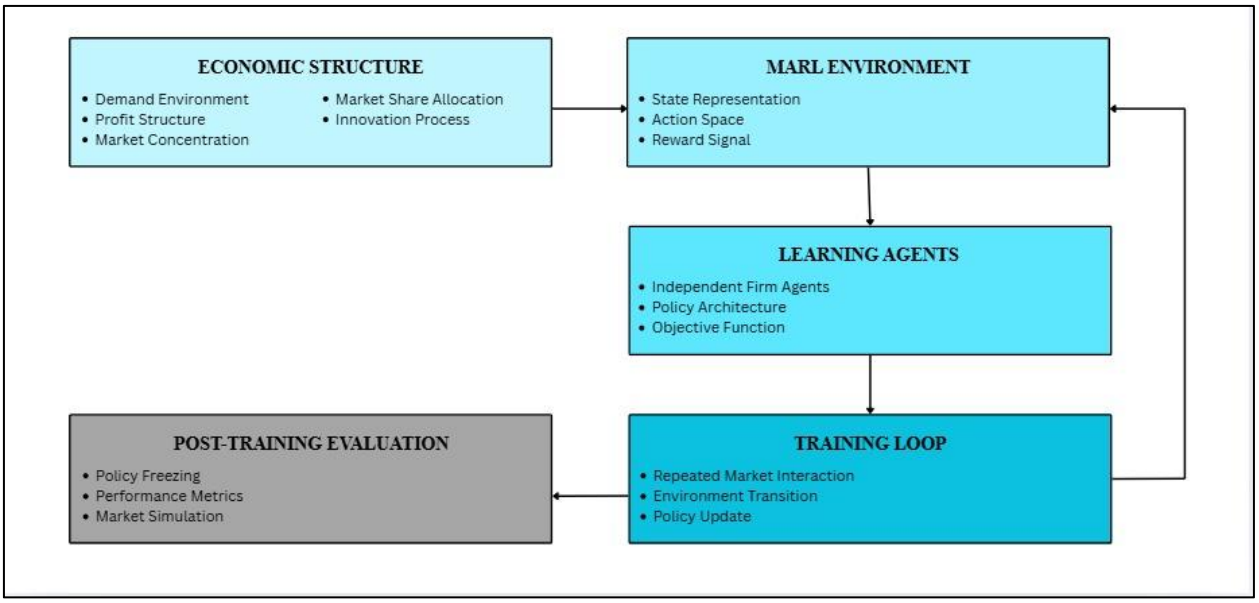

Figure 1: System Architecture

4.1. Environment Design

The industry demonstrates that the PettingZoo ParallelEnv framework could provide the significant empirical foundation for implementing multiple agents choosing their actions simultaneously within a shared environment. Moreover, the results may suggest that this reflects real-world oligopoly conditions, where firms set prices and innovation levels at the same time without seeing competitors' decisions in advance. Thus, findings might indicate the environment includes three fixed firms. However, data may show each period allows every firm to observe full market state. The state includes these vectors:

- All firms' prices
- Innovation stocks
- Market shares
- Marginal production costs
- Aggregate demand level
- Macroeconomic regime
- Substitute pressure
- Time index

The state space includes the current macroeconomic regime, allowing agents to condition pricing and innovation strategies on persistent boom-recession dynamics. Firms may well demonstrate that they possess significant empirical access to current market conditions, observing all critical developments unfolding across each period. Moreover, the important theoretical environment might indicate that strategic complexity persists throughout training, given that competitors' policies continue to evolve. However, rivals may show adaptive behavior rather than fixed responses. Thus, action space could appear continuously. In each period, firm (i) might select price and R&D investment subject to regulatory constraints:

$$\left(P_i(t),\ R_i(t)\right),$$

Given that continuous actions support theoretical realism, the model may suggest firms could adjust decisions gradually rather than committing large, fixed jumps. Furthermore, the important reward function might indicate that it corresponds exactly to the profit function in the economic model. Therefore, learning may show a direct focus on profits.

4.2. Learning Algorithm

Each firm is treated as an independent learning agent. We use an RL method called Proximal Policy Optimization (PPO), implemented through Stable-Baselines3. This method helps firms learn gradually through feedback and experience while keeping the decisions stable. Each firm decides prices and R&D investment levels. Firms do not coordinate by sharing information, and they are not programmed to do so. Each focus on maximizing its own profit in the long run.

During training firms compete against each other repeatedly in the same simulated market. Gradually, they adjust their strategies based on what works best. There is no central controller or supervised, hence pricing stability, innovation leadership, and market concentration comes naturally from competition. Thus, we can say that firms learn by competing again and again, and market outcomes develop from this ongoing adaptive interaction, not from solving a fixed theoretical equilibrium.

Table 1: Reinforcement Learning Configuration

Component	**Specification**
Algorithm	Proximal Policy Optimization (PPO)
Environment	PettingZoo ParallelEnv
Agents	3 independent learners
State Space	Fully observable
Action Space	Continuous (Price, R&D)
Episode Length	200 time steps
Training Horizon	1,000,000 steps
Discount Factor	0.99
Evaluation	Tournament simulation

4.3. Training Protocol

Each episode spans 200 periods, resulting in a long strategic time-period to allow:

- Accumulation of innovation stocks
- Reallocation of market shares
- Realization of macroeconomic regime shifts
- Propagation of supplier shocks

The discount factor is set to:

$$\delta = 0.99 ,$$

which induces strong intertemporal orientation and ensures that long-run innovation benefits meaningfully influence decision-making. Training proceeds for one million time steps to facilitate convergence of policy functions. Convergence is assessed through stabilization of:

- Average industry price
- Firm-level innovation investment
- Market shares
- Herfindahl-Hirschman Index (HHI)

Following training, policies are frozen and evaluated through repeated tournament simulations. In these evaluations:

- Agents no longer update their parameters
- Multiple independent stochastic runs are executed
- Outcome statistics are averaged across runs

This evaluation setup may well suggest that the significant empirical results could indicate stable, learned strategies rather than temporary behavior patterns observed during training. Moreover, the computational model might indicate that the structural industrial organization framework could transform into a decentralized, adaptive system. Thus, firms may show they avoid direct calculation of Nash equilibria. However, competition patterns might develop naturally through repeated interaction. Therefore, results could reflect realistic economic constraints. Given that this approach demonstrates that equilibrium forms as a dynamic learning process, the key theoretical findings may suggest that important methodological implications emerge naturally from the evidence. In light of these results, the study might indicate that treating equilibrium as a fixed, static solution could demonstrate significant limitations relative to adaptive approaches.

5. Results

We observed equilibrium patterns after policy convergence and subsequent tournament evaluation. All results are computed from static learned policies evaluated across multiple random simulations to ensure robustness. The analysis focusses on price, innovation, profitability, and resilience to shocks.

5.1. Pricing Outcomes

Figure 2 shows how the average industry price moves with marginal cost. After a short learning phase, the prices of firms converge toward a stable range above cost. This happens early and remains consistent across different simulation runs. Markups are moderate but positive, which fits normal price competition in differentiated markets. Firms can earn some margin, but competition prevents prices from rising too high. Prices also remain below the regulatory ceiling. There are no sudden joint price spikes or movements toward very high pricing. Instead, prices stabilize naturally without drifting into collusive levels. The outcome reflects normal competitive forces shaped by demand sensitivity, innovation differences, and regulatory limits.

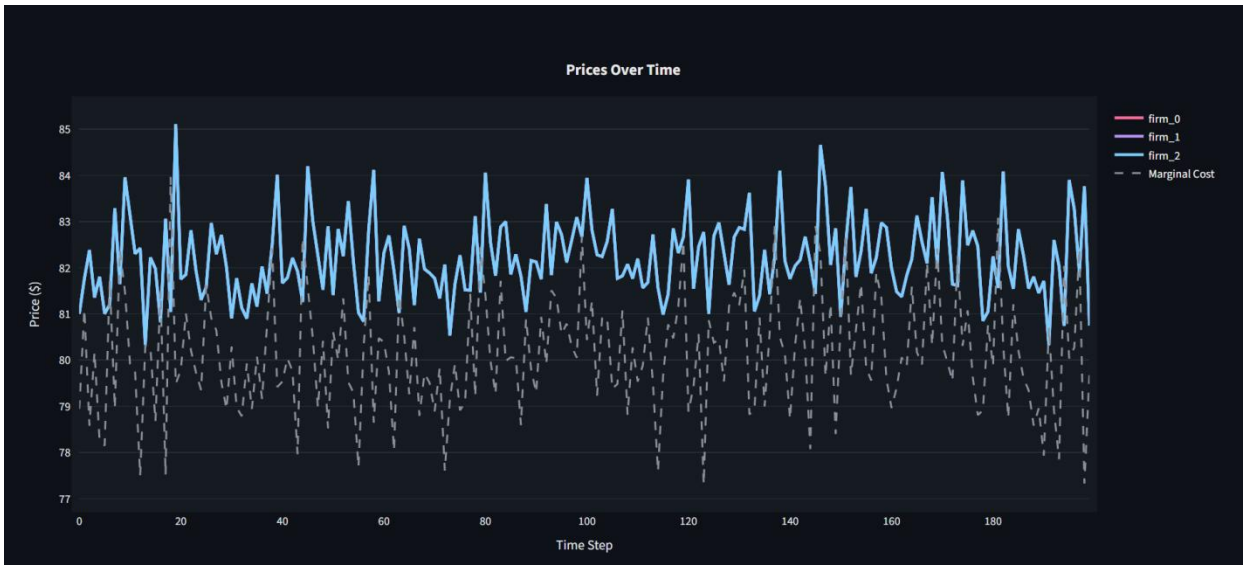

Figure 2: Evolution of prices over time

5.2. Innovation and Market Structure

Figure 3 shows the stock trajectories at the firm level. A firm emerges as a innovation leader, accumulating a substantially larger stock over time. Since innovation evolves cumulatively over time, early investment helps compound, generating path dependence consistent with stepped innovation models.

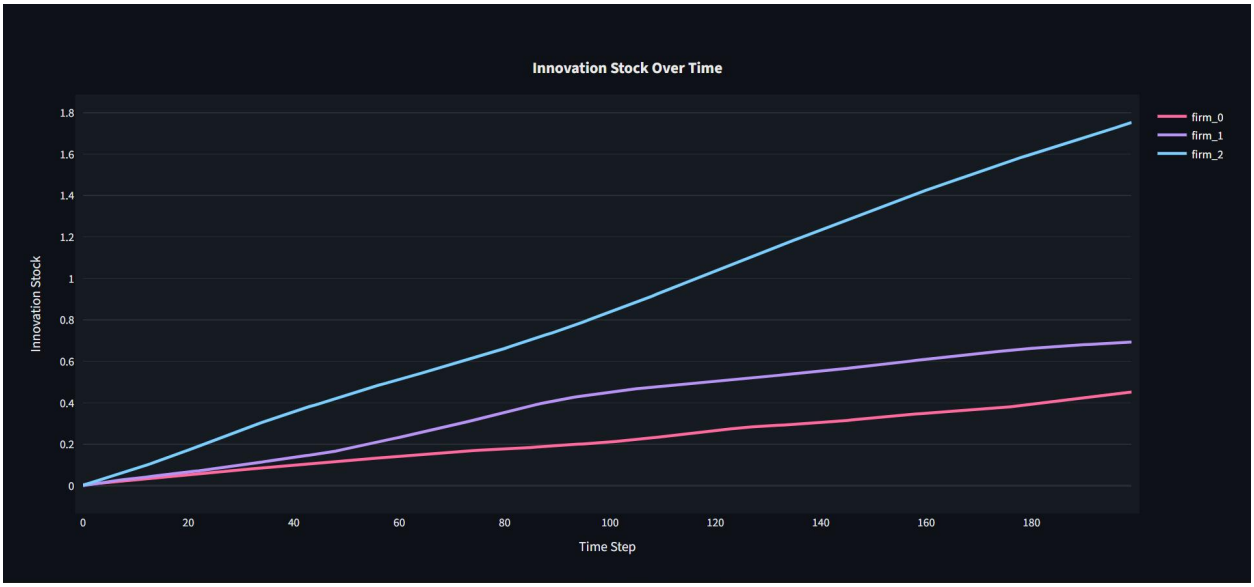

Figure 3: Innovation stock trajectories over time

Figure 4 shows the relationship between each firms' innovation stock and market share. Firms that build higher levels of innovation gather large shares of total demand. The leading firm invests more in innovation over time and dominates the market. This relationship shows that success comes from innovation-based differentiation built into the demand system. Market leadership is driven by long-term capability investment, not by aggressive price cutting.

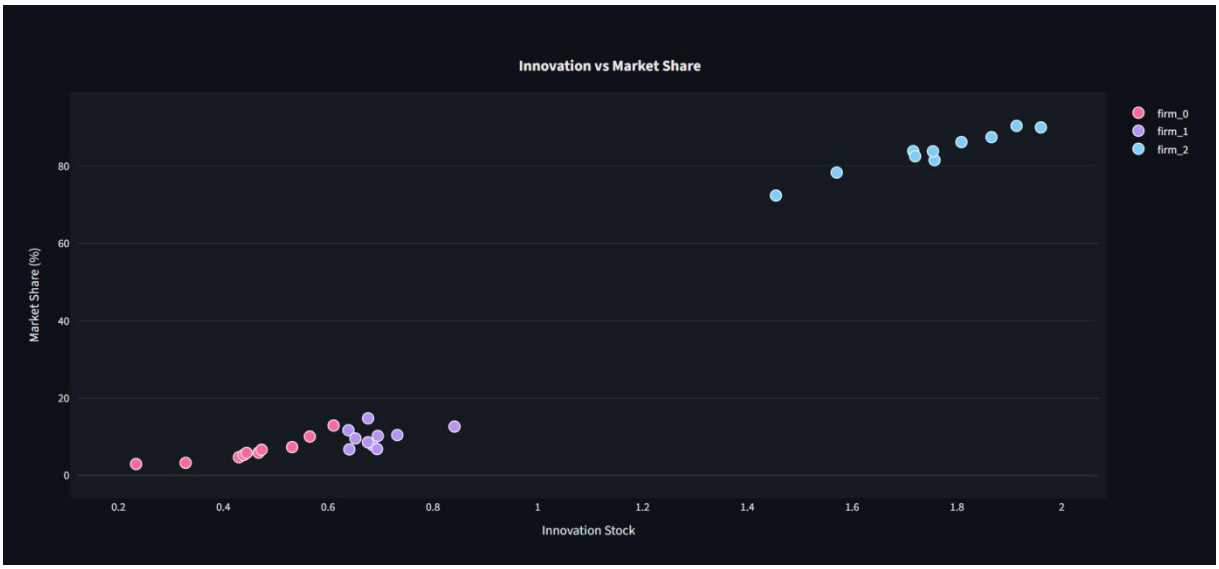

Figure 4: Innovation stock and corresponding market shares across firms

Figure 5 shows the final equilibrium market structure. One firm gathers about 83.6% of total demand, while the others (rivals) gather about 9.9% and 6.5% respectively. The structure does not show full monopoly. Entry barriers, regulatory constraints, and diminishing returns to innovation to prevent market foreclosure. The final concentration reflects cumulative investment dynamics rather than predatory exclusion

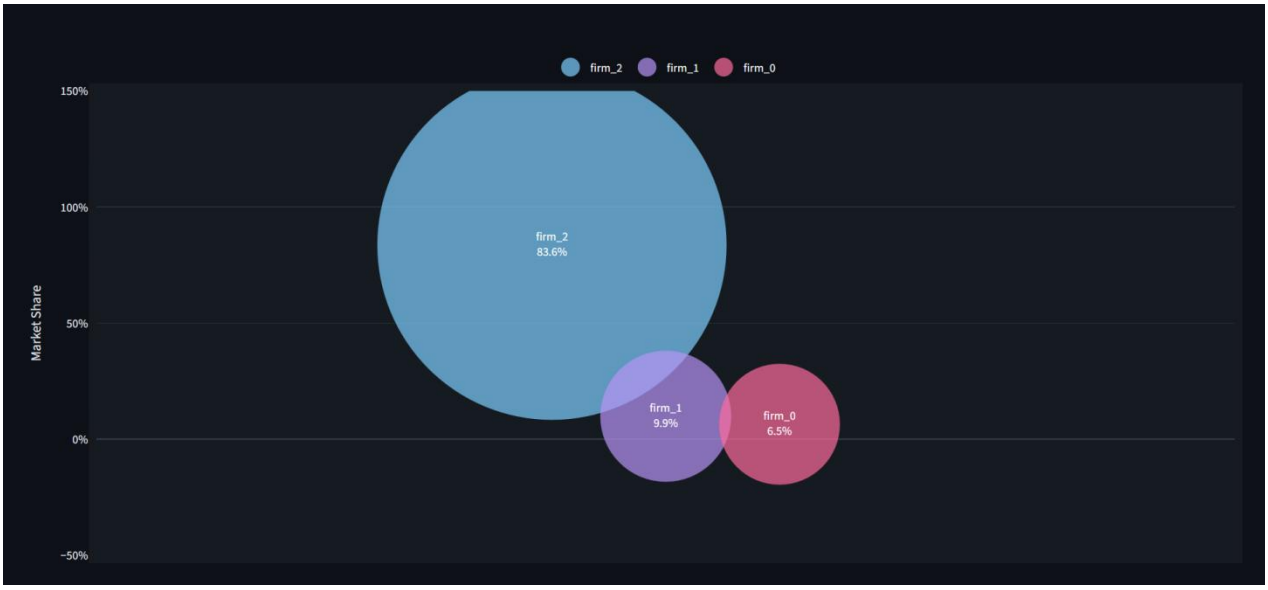

Figure 5: Final equilibrium market share distribution

Figure 6 portrays the HHI. Market concentration increases steadily as the degree of innovation grows, and then settles at a high level without reaching monopoly. Hence the market becomes highly concentrated but not completely dominated by one firm. This concentration arises from continuous R&D investment. The pattern shows how long-term investment and sunk costs can create lasting differences between firms.

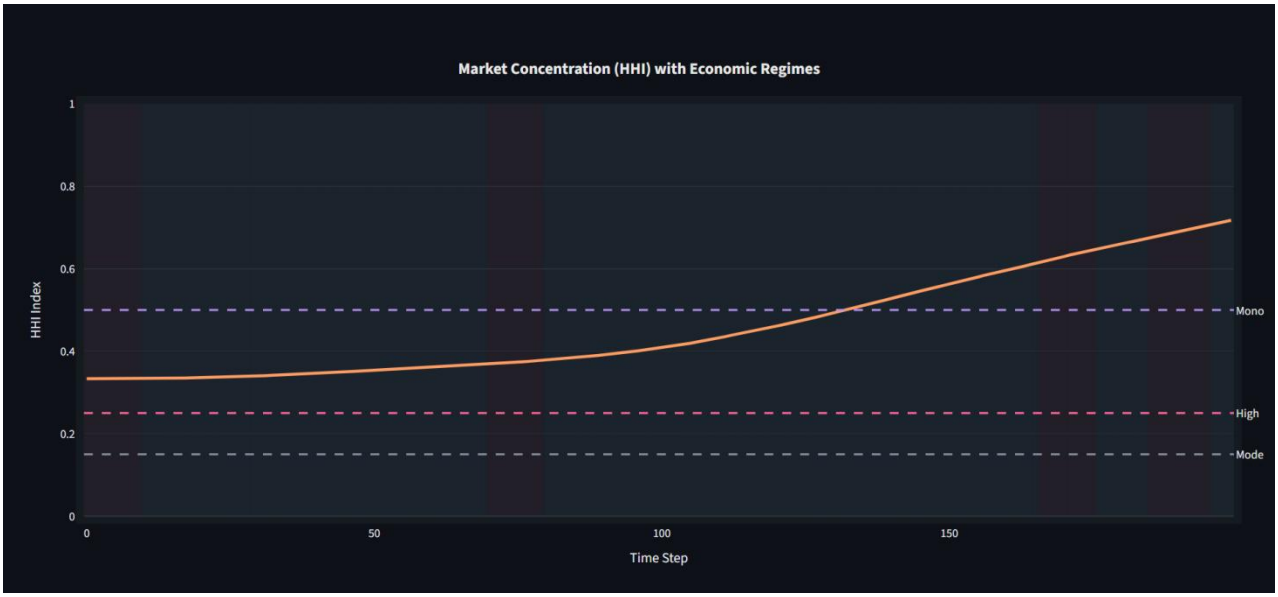

Figure 6: HHI visualization

5.3. Profitibility and Stability

Figure 7 shows the cumulative profits by firms over time. All firms remain profitable in long-run equilibrium, including those with smaller market shares. Differentiated demand and pricing limits prevent margins from collapsing, while the leader earns higher profits. After convergence, the system becomes stable. Prices move slightly with cost changes; market shares adjust gradually, and innovation spending levels off at steady levels. There are no price wars or unstable swings. The firms settle into stable, self-sustaining competitive equilibrium.

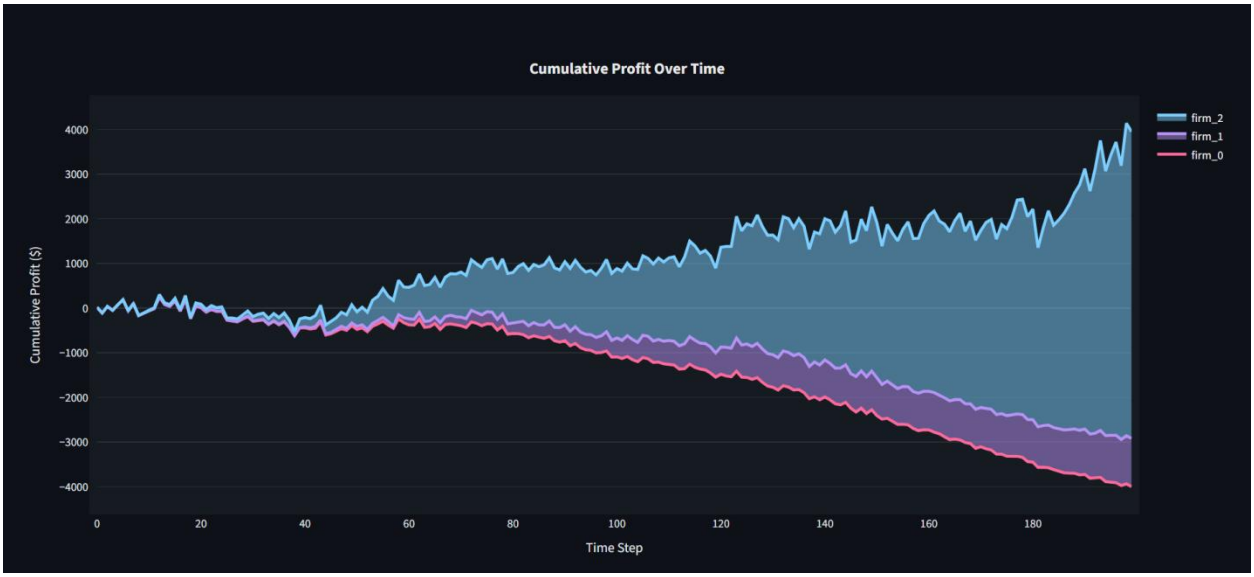

Figure 7: Cumulative profit of firms over time

5.4. Robustness to Macroeconomic and Cost Shocks

The model integrates random macroeconomic regime shifts and supplier cost volatility. After the cost fluctuations, equilibrium prices remain withing the stable markup band. Firms learn and adjust prices in sync with the marginal cost changes, rather than showing amplified responses. Innovation or uniqueness adjusts gradually, reflecting its role in long-term strategy. Market concentration patterns remain qualitatively stable across the simulated demands, showing structural robustness.

Table 2: Equilibrium market outcomes

Metric	Firm 0	Firm 1	Firm 2	Industry
Final Market Share (%)	6.5	9.9	83.6	100
Average Price ($)	~82	~82	~83	~82
Avg. Marginal Cost ($)	~80	~80	~80	~80
Avg. Markup (%)	~2-3%	~2-3%	~3-4%	-
Avg. Episode Profit ($)	921	1115	5989	-
HHI	-	-	-	0.708

6. Implications

The results generate implications for strategic management, competition policy, and computational industrail organization. This is because competitive outcomes emerge from adaptive interaction within a structurally specified oligopoly model; the implications reflect economically grounded dynamics rather than stylized equilibrium assumptions.

6.1. Managerial Implications

Long-term success could indicate that continuous innovation, rather than price reduction, may well suggest the more significant empirical path toward sustainable competitive advantage. Moreover, the findings demonstrate that firms investing consistently in research and development may build stronger theoretical capabilities over time. Thus, price reductions might affect sales briefly. However, evidence may show lasting advantage requires quality differentiation. Considering the data, strong capabilities could indicate innovation drives performance over short-term price tactics.

Furthermore, the important evidence suggests that regulation might reasonably indicate a balanced approach—limiting excessive pricing without constraining innovative activity. Given that price ceilings and compliance costs remain present, the significant results could demonstrate that firms still invest in R&D and remain profitable. Therefore, moderate regulation may show it controls markups while allowing progress. Notwithstanding this result, high fixed costs might increase concentration. Thus, findings may show larger firms could strengthen position over time.

The key evidence could indicate that stable prices do not automatically suggest collusion among competing firms. Additionally, the significant empirical findings demonstrate that prices settle slightly above cost without rising to extreme levels. However, simple markets may show learning algorithms affect collusion. In light of these results, differentiated markets might indicate pricing stability results from normal competitive forces. Therefore, price discipline may reflect structure, not coordination.

6.2. Policy Implications

High market concentration in the model (HHI ≈ 0.708) could indicate that one firm builds a significant innovation advantage over time-not because firms block competitors or coordinate on prices. Moreover, this important difference may suggest that antitrust policy must examine the evidence carefully. A company becoming dominant due to better technology and higher productivity might indicate something fundamentally different from firms secretly working together to limit competition. Thus, policymakers may show awareness of market power arising from innovation versus coordination. Therefore, regulators could separate power from collusion.

Concerns about algorithmic pricing may also depend on market structure. Given that the findings demonstrate learning algorithms can lead to collusion in very simple market settings (Calvano et al., 2020), the significant empirical evidence could indicate that context matters considerably for the results. However, findings may suggest that in more realistic markets-where products differ, firms invest in innovation, and regulations limit pricing-collusion does not automatically occur (Ezrachi et al., 2016). Furthermore, results may show that regulators should evaluate specific industry structure. Nevertheless, assuming pricing algorithms always lead to collusion might affect policy outcomes. The recession and boom cycles alter the demand levels but do not overturn innovation leadership. Structural heterogeneity persists across regimes, consistent with dynamic industry models (Ericson et al., 1995).

6.3. Research Implications

Decentralized reinforcement learning could indicate that a significant tractable alternative to solving high-dimensional dynamic equilibria exists (Pakes et al., 2001). Moreover, the important evidence may suggest that embedding structural demand, significant innovation dynamics, and critical regulatory frictions within an adaptive learning framework demonstrates that the model preserves key economic interpretability while allowing competitive patterns to emerge endogenously (Zhang et al., 2021). Thus, findings might support modeling dynamic competition as an adaptive process. However, concentration and pricing discipline may affect technological leadership through repeated interaction rather than static equilibrium. Considering these results, findings could show strategy theory and industrial organization bridge through this approach.

7. Conclusion

This study indicates that when firms learn and adapt within a well-defined oligopolistic framework, significant empirical findings suggest that resulting market patterns are closely similar to those outlined in industrial organization theory. Important evidence could indicate that pricing remains disciplined as significant empirical research demonstrates that concentration develops gradually with time. Moreover, the findings may suggest that leadership shows motivation through continuous innovation. Thus, results may show outcomes are not incorporated into models from the start. Therefore, evidence could suggest equilibrium emerges through adaptive behavior rather than imposed advance conditions.

Considering the significant theoretical findings, differentiated price competition paired with cumulative innovation may well suggest that ensuing industry dynamics closely match important observable market patterns. The results might indicate that firms prefer generally constant pricing ranges with positive markups, whereas prolonged R&D expenditure could demonstrate that lasting variations in market share emerge naturally. However, market concentration may show it occurs mostly from gradual buildup of skills rather than exclusionary pricing. Thus, regulatory limits might affect competitive behavior while preserving incentives for innovation.

Given that the key evidence demonstrates that industry outcomes may well reflect the combined impact of continuing competence development, the institutional context in which businesses operate could indicate that significant methodological findings matter substantially. Furthermore, this critical framework might suggest that it combines two important ways of thinking about markets that are frequently kept separate. Additionally, results may show rivalry and regulatory costs are embedded in context where businesses function. However, the model could

support clear economic reasoning while avoiding predetermined results. Thus, market patterns might emerge as enterprises interact over time, making findings suggest this framework could provide a useful setting for investigating strategy and policy issues.

Given that significant empirical evidence could indicate that the simulator code has been made available to support important research purposes and further meaningful transparency, this contribution may well suggest a critical methodological advance. Moreover, the significant tournament datasets may suggest that they are saved in a re3data-compliant repository, with all model parameters thoroughly documented. Thus, findings might show other researchers can reproduce results or adjust the framework. Additionally, evidence may indicate that the framework could broaden the theoretical scope. Considering the findings demonstrating that RL agents function in a designed economic context, the important empirical results could plausibly indicate that they may recreate key elements of real-world marketplaces, establishing significant theoretical and practical implications for the study. Therefore, significant evidence may suggest that the framework demonstrates that it provides important reproducible computational foundations for future research. However, results might indicate that the framework could support strategic and industrial study. Thus, findings may show that this approach supports future research. Nevertheless, results could indicate that broader applications may exist.

8. Future Scope

The next stage of development may well suggest that expanding the significant empirical framework from a fixed-structure oligopoly into a fully dynamic industry evolution model could plausibly demonstrate important theoretical implications for how the critical methodological approach is applied (Ericson & Pakes, 1995). Moreover, a primary extension involves endogenous entry and exit, where the findings may suggest that firms enter if expected discounted profits exceed a sunk entry cost and exit when cumulative profits fall below a solvency threshold (Ericson & Pakes, 1995). Furthermore, market structure might indicate firms respond to profitability conditions. Thus, results may show structure evolves as conditions shift (Aghion et al., 2001).

$$E[V_i] - F_{entry} \geq 0,$$

(equation 15)

Allowing the number of active firms to change over time lets us study creative destruction, industry shakeouts, and barriers to entry caused by innovation, like dynamic industry models (Ericson & Pakes, 1995; Aghion et al., 2001). Incomplete and uneven information, like private cost shocks, demand uncertainty, and unclear signals about competitor innovation, can make random innovation processes more realistic (Ericson & Pakes, 1995). These changes would enable the examination of strategic uncertainty, signaling, and leapfrogging dynamics in situations with poor monitoring (Aghion et al., 2001). This would strengthen the model's link to real-world industrial settings.

At present, the framework helps us understand how markets change and how companies adjust their strategies. However, it could be even more useful. It might create a safe space to test company tactics and government laws before putting them into practice. By including both human managers and AI agents, we can compare human decision-making with algorithm-driven behavior and learn more about each one's strengths (Calvano et al., 2020). Adding elements like antitrust enforcement, merger testing, innovation incentives, and price control would turn the system into a virtual lab for experimenting with different market interventions (Calvano et al., 2020).

The model may suggest that future research could more clearly demonstrate that specific stakeholder groups might benefit from significant market outcomes. However, rather than examining only the overall aggregate results, the analysis should separately indicate how consumers may gain important advantages, how businesses could generate substantial profits, and how innovation might create critical long-term value (Aghion et al., 2001). The framework

would need to carefully look at the demand factors, such as behavior of customers to price changes and the effect of innovation on various industries (Berry et al., 1995). Therefore, the model will feel dependable and reliable. Nevertheless, with these improvements, the framework could become a practical tool for understanding how markets change over time.

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